

Solution of MGMT 670 Midterm Exam (Version A)

Fall 2025

Name (Print): _____

Student ID No.: _____

EXAM Rules:

This is an open-book, open-notes exam.

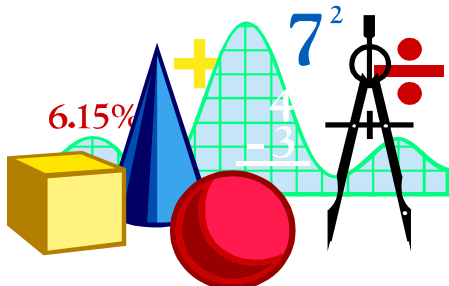
No cellphone. No internet. No cheating!

PART I. MULTIPLE CHOICES

1. Circle your answer on the exam.

PART II. WORK PROBLEMS

1. Give neat, clean, and well organized answers.
2. Show your work to support your answers.



Part I: Multiple Choices (10 Questions, 30 Points Total)

Identify the letter of the choice that **best** completes the statement or answers the question.

1. Which of the following is true about correlation coefficient?
 - a. Correlation coefficient lies within the interval $[0, 1]$.
 - b. If the data lie on a straight line, the correlation coefficient is 1.
 - c. A correlation coefficient of 0 implies no relationship between the two variables
 - d. None of the above is correct.
2. A delivery company tracks the number of packages delivered per day across its distribution centers. The mean daily deliveries are 1,200, with a standard deviation of 300. Consider the two scenarios:
Scenario A: Assume the distribution of deliveries is approximately normal.
Scenario B: Assume the distribution of deliveries is not necessarily normal.
Using the interval 600 to 1,800 packages per day, which statement is correct?
 - a. Under Scenario A, about 99% of centers are in this range.
 - b. Under Scenario A, about 68% of centers are in this range.
 - c. Under Scenario B, at least 75% of centers are in this range.
 - d. Under Scenario B, at least 89% of centers are in this range.
3. A basketball player has an 80% success rate for free throws. She attempts 5 free throws in a game. Let X represent the number of free throws she **misses**. Which of the following statements about X is true?
 - a. $P(X \geq 1) < P(X \leq 1)$
 - b. $P(X \geq 1) = P(X \leq 1)$
 - c. $P(X \geq 1) > P(X \leq 1)$
 - d. Cannot be determined from the statement of the problem.
4. A quality inspector classifies 200 light bulbs as defective or non-defective. The results are shown below. What is the probability that a randomly chosen bulb is classified **the same** by the inspector and the supervisor?

	Supervisor: Non-defective	Supervisor: Defective
Inspector: Non-defective	90	15
Inspector: Defective	20	75

 - a. Less than 50%
 - b. Between 50% and 70%
 - c. Between 70% and 85%
 - d. More than 85%
5. A survey of 1,000 people estimates that 70% prefer online shopping. The margin of error for a 95% confidence interval is 4%. Which of the following changes would reduce the margin of error?
 - a. Decrease the sample size
 - b. Use a smaller estimated proportion p
 - c. Lower the confidence level to 90%
 - d. None of the above

6. A checkout scanner makes an error 4% of the time, independent of the item price. Half the errors increase the price by \$2, and half decrease it by \$1. What would be the impact on the expected price of the items sold?
- The expected price will not change
 - The expected price will go up
 - The expected price will go down
 - Cannot be determined from the statement of the problem
7. You are considering two investment options: the S&P 500 and Core Bonds. Historical data shows: S&P 500: Expected return = 10%, Standard deviation = 8%; Core Bonds: Expected return = 2%, Standard deviation = 2%. Correlation coefficient between them is **-0.3**. Two portfolios are considered:
Portfolio A: 80% S&P 500, 20% Bonds
Portfolio B: 50% S&P 500, 50% Bonds
Which of the following statements is NOT correct? (round to 1 decimal)
- Expected return of Portfolio A is 8.4%
 - Expected return of Portfolio B is 6.0%
 - Standard deviation of Portfolio A is 38.0%
 - Standard deviation of Portfolio B is 3.8%
8. A cybersecurity firm monitors login attempts to detect suspicious activity. From past data, 4% of all login attempts are from hackers. The system also found that: If an attempt is from a hacker, there is a 90% chance it will trigger a security alert. If an attempt is from a legitimate user, there is a 5% chance it will (falsely) trigger a security alert. Given that a security alert was triggered, what is the probability that the login attempt was from a hacker? (Round to 3 decimals)
- 0.036
 - 0.084
 - 0.429
 - 0.571
9. A company inspects cars from four rental agencies. The fleet mean score is 50 with standard deviation 80. A sample of 64 cars is checked. What is the approximate probability that the **sample mean** exceeds 60?
- Less than 10%
 - Between 10% and 30%
 - Between 30% and 50%
 - Greater than 50%
10. A teacher records exam scores for 200 students. The distribution of scores is **left-skewed**. She randomly selects 50 students to estimate the average exam score and calculates a 95% confidence interval for the mean: (70, 90). Which of the following statements is correct?
- The sample mean is likely larger than the median.
 - The sampling distribution of the sample mean is approximately normal, even though the population distribution is left-skewed.
 - 95% of students' exam scores fall between 70 and 90.
 - There is a 95% probability that the population mean falls within this interval.

Part II: Show Your Work (2 Problems, 20 Points Total)

Problem 1

The flight time from Chicago O'Hare (ORD) to Purdue University (LAF) is normally distributed with a **mean of 60 minutes** and a **standard deviation of 5 minutes**.

a. Flights with durations longer than **66 minutes** are considered “delayed.” What percentage of flights are delayed? (round to 3 decimals) (2 pts)

b. A random sample of 50 such flights is selected. Let $\bar{p}$ be the sample proportion of delayed flights in this sample. Calculate the expected value and standard deviation of $\bar{p}$. (round to 3 decimals) (2 pts)

c. Using the normal approximation to the binomial, what is the probability that, in a sample of 50 flights, **at most 10 flights** are delayed? (round to 3 decimals) (2 pts)

d. For a random sample of 25 flights, specify the sampling distribution of the sample mean flight time. What is the probability that the sample mean flight time is **between 59 and 61 minutes**? (round to 3 decimals) (2 pts)

e. Continuing from part d, suppose the sample of 25 flights has a sample mean of 61 minutes. Construct a 95% confidence interval for the mean flight time. (round to 2 decimals) (2 pts)

Problem 2

McHuffer Condominiums of Florida is trying to estimate the size of housing development they should plan for next to the Gulf of Mexico. Due to economic uncertainty, it is hard to predict demand. The management has prepared the following **profit payoff** table (in millions of dollars):

Payoff in millions	Demand	
Size of development	Low	High
Small	2	3
Large	1	5
Probability	0.6	0.4

- a. Find the expected payoff for the two decision alternatives. What is the best decision based on the expected value criterion? (3 pt)

- b. What is the expected value of perfect information? (2 pt)

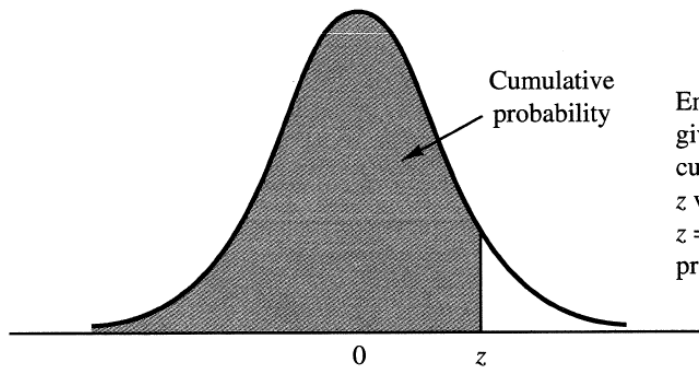
McHuffer is considering a survey to help evaluate the demand for the condominium development. The survey predicts either **favorable** or **unfavorable** response. Based on past results:

- $P(\text{Favorable} \mid \text{High Demand}) = 0.7$
- $P(\text{Favorable} \mid \text{Low Demand}) = 0.2$

c. What is the expected value of sample information? (3 pts)

d. If the survey is unfavorable, should McHuffer build a small sized development or a large sized development? Support your argument with numbers. (2 pts)

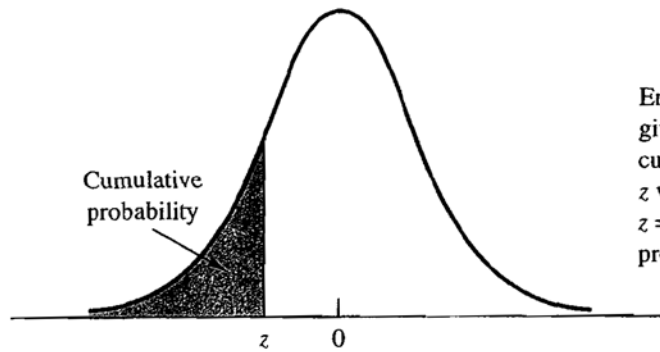
CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in the table give the area under the curve to the left of the z value. For example, for $z = 1.25$, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in this table give the area under the curve to the left of the z value. For example, for $z = -.85$, the cumulative probability is .1977.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Part I Solutions (Version A)

1. D Correlation ranges from -1 to 1, not [0,1]. If data lie on a straight line with negative slope, correlation is -1. Correlation of 0 indicates no linear relationship, but there could be a nonlinear relationship.
2. C Scenario B: Distribution not necessarily normal $\rightarrow$ use Chebyshev's inequality.
 $k = 2 \rightarrow$ minimum proportion $\geq 1 - 1/k^2 = 1 - 1/4 = 0.75$. For scenario A, it's 95%.
3. A $X = \#$ of misses, $P(\text{miss}) = 0.2$, 5 attempts.
 $P(X = 0) = 0.8^5 \approx 0.328$
 $P(X \leq 1) = P(X=0) + P(X=1) \approx 0.328 + 0.41 = 0.738$
 $P(X \geq 1) = 1 - P(X=0) \approx 0.672 \rightarrow P(X \geq 1) < P(X \leq 1)$
4. C Agreement = $(90+75)/200 = 165/200 = 0.825 \approx 82.5\%$
5. C Margin of error = $z \times \sqrt{p(1-p)/n}$. Decreasing confidence level $\rightarrow$ smaller $z \rightarrow$ smaller margin of error.
6. B Expected change per error = $0.5 \times 2 + 0.5 \times (-1) = 0.5$.
Error probability = 0.04 $\rightarrow$ Expected change per item = $0.04 \times 0.5 = 0.02$
Positive $\rightarrow$ expected price increases
7. C Expected return of Portfolio A = 8.4%
Expected return of Portfolio B = 6%
Standard deviation of Portfolio A = 6.3%
Standard deviation of Portfolio B = 3.8%.
8. C Bayes' theorem $P(H|A) = P(A|H)P(H) / [P(A|H)P(H) + P(A|L)P(L)]$
 $= 0.9 \times 0.04 / (0.9 \times 0.04 + 0.05 \times 0.96) = 0.036 / (0.036 + 0.048) \approx 0.429$
9. B Standard error = $80/\sqrt{64} = 10$
 $Z = (60-50)/10 = 1 \rightarrow P(Z > 1) \approx 0.1587 \approx 16\%$
10. B Left-skewed distribution $\rightarrow$ mean < median (tail on the left).
By CLT, if sample size is reasonably large ($n \geq 30$), the distribution of the sample mean is approximately normal, regardless of the population distribution.
C is wrong because confidence interval refers to the population mean, not individual student scores. Individual scores can be outside the interval.
D is wrong because once the interval is calculated, the population mean is fixed. Probability doesn't apply to this single interval.
Correct interpretation: "We are 95% confident that this interval contains the true mean." or If we repeatedly take random samples of the same size ($n = 50$) from the population and calculate a 95% CI for each sample, about 95% of these intervals would contain the true population mean, that is, "In repeated sampling, 95% of intervals would contain the true mean."

Part II Solutions

Problem 1

- a. Flights delayed ($X > 66$): $Z = (66-60)/5 = 1.2$, $P(X > 66) = 1 - 0.8849 \approx 0.1151 \rightarrow 0.115$.
- b. Sample proportion $\bar{p}$: $E[\bar{p}] = 0.115$, $SD[\bar{p}] = \sqrt{(0.115 \cdot 0.885/50)} \approx 0.045$.
At most 10 delayed flights: $\mu = np = 50 \cdot 0.115 = 5.75$, $\sigma = \sqrt{np(1-p)} \approx 2.256$, $Z = (10.5 - 5.75)/2.256 \approx 2.11$, $P(Y \leq 10) \approx 0.9826 \rightarrow 0.983$.
- c. It's OK if students did not use 0.5 correction. In that case, $Z = (10 - 5.75)/2.256 = 1.88$, and probability is 0.970. Or students may use $P(\bar{p} \leq 0.2)$ to compute, which leads to $Z = 1.89$ and the probability is 0.9706.
- d. Sampling distribution of sample mean flight time: $\bar{X} \sim N(\mu=60, \sigma_{\bar{X}}=1)$,
 $Z_1 = (59-60)/1 = -1$, $Z_2 = (61-60)/1 = 1$, $P(59 \leq \bar{X} \leq 61) = P(-1 \leq Z \leq 1) \approx 0.6826 \rightarrow 68.3\%$
- e. 95% CI for $\bar{X} = 61$: $SE = 1$, $CI = 61 \pm 1.96 \cdot 1 \approx (59.04, 62.96)$;

Problem 2

(a) Expected Payoff: $E[\text{Payoff}] = P(\text{Low}) * \text{Payoff}(\text{Low}) + P(\text{High}) * \text{Payoff}(\text{High})$

Small: $E[\text{Small}] = 0.6 * 2 + 0.4 * 3 = 1.2 + 1.2 = 2.4$

Large: $E[\text{Large}] = 0.6 * 1 + 0.4 * 5 = 0.6 + 2 = 2.6$

Decision: Build Large development (higher expected payoff $2.6 > 2.4$).

(b) $EVPI = E[\text{Payoff with perfect information}] - \text{Expected payoff}$

If Low demand: choose Small $\rightarrow$ payoff = 2

If High demand: choose Large $\rightarrow$ payoff = 5

$E[\text{Payoff with perfect information}] = 0.6 * 2 + 0.4 * 5 = 1.2 + 2 = 3.2$

EVPI = 3.2 - 2.6 = 0.6

(c) EVSI:

Step 1: Compute $P(\text{Fav})$ and $P(\text{Unfav})$

$P(\text{Fav}) = 0.7 * 0.4 + 0.2 * 0.6 = 0.28 + 0.12 = 0.40$ and $P(\text{Unfav}) = 1 - 0.40 = 0.60$

Step 2: Posterior probabilities

$P(\text{High} | \text{Fav}) = (0.7 * 0.4) / 0.40 = 0.7$ and hence $P(\text{Low} | \text{Fav}) = 0.3$

$P(\text{High} | \text{Unfav}) = (0.3 * 0.4) / 0.60 = 0.2$ and hence $P(\text{Low} | \text{Unfav}) = 0.8$

Step 3: Expected payoff given survey

Fav: Small = $0.32 + 0.73 = 2.7$, Large = $0.31 + 0.75 = 3.8 \rightarrow$ choose Large

Unfav: Small = $0.82 + 0.23 = 2.2$, Large = $0.81 + 0.25 = 1.8 \rightarrow$ choose Small

Step 4: Expected value with sample information

$E[\text{With survey}] = 0.4 * 3.8 + 0.6 * 2.2 = 1.52 + 1.32 = 2.84$

EVSI = 2.84 - 2.6 = 0.24

(d) Decision if Survey is Unfavorable

Posterior expected payoff: Small = 2.2, Large = 1.8

Decision: Build Small development ($2.2 > 1.8$)